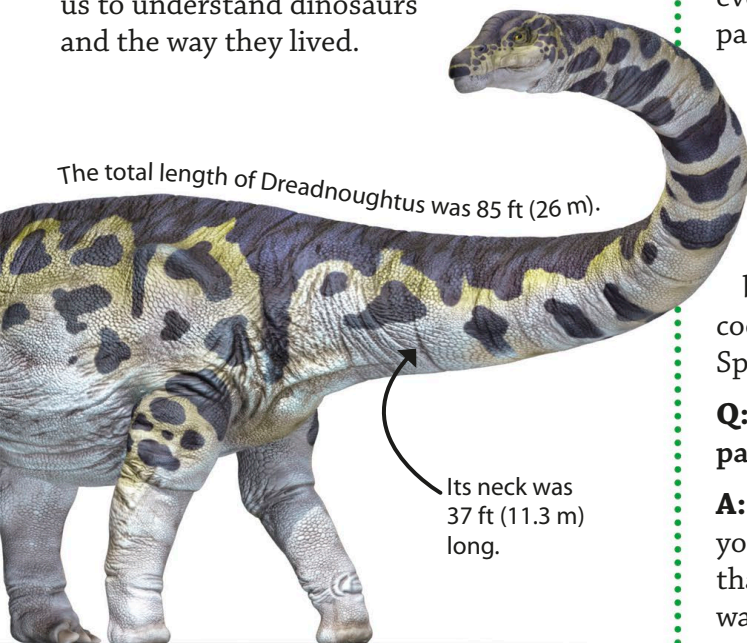


Q: How do you know where to look for dinosaur fossils?

A: First, find rocks of the right age. Dinosaurs lived from 235 to 66 million years ago, during the Mesozoic Era. Fossils are only found in rocks formed by sediments, such as mud. Don't bother looking for fossils in volcanic rocks or in rocks that have been heated and squeezed by the Earth's crust. They won't be there! Finally, look in deserts, where erosion is constantly uncovering new bones. If you find those three things, walk, keep your eyes on the ground, and be patient. You will almost always find fossils.

Q: What sort of tools do you use?

A: In the field, we use simple tools, such as pickaxes, shovels, hammers, and chisels. In the laboratory, our tools are high-tech. Robotics, 3-D scanners, 3-D printers, and medical equipment all help us to understand dinosaurs and the way they lived.



The total length of *Dreadnoughtus* was 85 ft (26 m).

Its neck was 37 ft (11.3 m) long.



Dr. Lacovara stands next to the curving neck bones.

Digging up Dreadnoughtus

It took the team five years to dig up the remains of the super-massive Dreadnoughtus. More than 70 percent of its bones were found.

Q: How do you know what dinosaurs looked like in real life?

A: We know their shape from how their bones fit together. Scars on the bones show us where muscles were. Sometimes armor plates, spines, and spikes are preserved and even scales and feathers. To understand parts that do not fossilize, like lungs and eyes, we look at birds, which are descended from dinosaurs, and also crocodiles.

Q: Do you have a favorite dinosaur?

A: Dreadnoughtus! After spending 10 years digging up and studying it, it has become a member of my family. For sheer coolness, though, nothing beats Spinosaurus.

Q: What do you love most about paleontology?

A: With only a shovel, paleontology allows you to "time-travel" to ancient landscapes that existed before humans, when the world was ruled by Nature alone.